

10 Assignment Questions – Time Series & Forecasting

1. What is Time Series Analysis? Explain its importance and list different components of a time series with suitable examples.
2. Explain Simple, Double, and Triple Exponential Smoothing methods. How do level, trend, and seasonality affect forecasting?
3. Describe the steps involved in visualizing a time series. How do decomposition and moving averages help in understanding the underlying patterns?
4. What is stationarity in time series? Explain any three methods to check stationarity and describe how non-stationary data can be made stationary.
5. Define AR(p) and MA(q) models. Briefly explain how ACF and PACF plots help in identifying suitable AR and MA orders.
6. What is an ARIMA model? Explain the role of the parameters (p, d, q) and describe how the backward shift operator is used in ARIMA modeling.
7. Explain the concept of seasonality in a time series. What is SARIMA? Describe the role of seasonal parameters (P, D, Q) with an example.
8. Differentiate between ARIMA and SARIMA models. In what situations is SARIMA preferred over ARIMA?
9. What is the purpose of simulating time series data? Write short notes on simulating AR, MA, and ARMA processes.
10. Describe the steps of a complete Time Series Forecasting workflow—from visualization, stationarity testing, model selection, model fitting, to forecasting and evaluation.